Model Evaluation — ML-Based Traffic Speed Prediction

This document describes the evaluation of the trained models published in this repository (`paper_artifacts/trained_models_base` accompanying the paper "Cloud-Fog-Edge Deployment of ML-Based Traffic Speed Prediction: An Empirical Analysis of Performance, Cost, and Scalability".

Models

One multilayer perceptron per sensor direction (37 sensor directions; layer sizes 96-256-256-96-256, ReLU with dropout, ~142k parameters, RMSprop optimiser, MSE loss, float32). Each model ships with its fitted input (x) and output (y) RobustScaler as `.pkl` files. Models were trained on the initial three-year window (1 December 2020 to 1 December 2023) of five-minute average-speed records from Greater Manchester drakewell sensors, using 15 engineered features (cyclic time encodings and event flags). These are the base models from which all warm-start re-training scenarios in the paper are initialised.

Evaluation protocol

Each sensor's data is split chronologically 80/20; the final 20% of the training window serves as the held-out test set (approximately the last seven months of 2023 for the three-year window). Metrics: mean absolute error (MAE, mph), root mean squared error (RMSE, mph), and mean absolute percentage error (MAPE, %). For context, each row also reports two properties of the sensor's full speed series: the coefficient of variation of speed over the full series (Speed CoV, dimensionless); the within-time-slot CoV (standard deviation of speed after removing the sensor's time-of-week mean profile, divided by mean speed, computed on stable 2023 data); and the congestion frequency (share of five-minute intervals below 50% of the sensor's 85th-percentile free-flow speed). A complementary common-test-set evaluation (identical held-out periods for all training scenarios) is summarised in the last section.

Per-sensor results — 3-year training regime

All rows in this table come from the **3-year initial training regime** (trained on 1 December 2020 to 1 December 2023); the held-out test set is the final 20% of that window, i.e. approximately mid-2023 to December 2023 (exact boundary varies slightly per sensor with data completeness). These are the published models in `trained_models_base3y/`.

#	Sensor	MAE (mph)	RMSE (mph)	MAPE (%)	Speed CoV	Within-slot CoV	Congestion (%)	Test samples
1	1020 nw	1.95	2.92	8.02	0.128	0.099	0.63	61,132
2	1020 se	1.67	2.42	6.48	0.139	0.092	0.41	60,958
3	1056 e	1.36	1.92	6.06	0.086	0.078	0.11	59,752
4	1056 w	1.75	2.47	7.57	0.099	0.087	0.17	60,544
5	1063 e	1.45	2.50	5.76	0.097	0.082	0.43	61,213
6	1063 w	1.59	2.47	5.50	0.110	0.076	0.52	61,253
7	1071 ne	2.09	3.65	10.48	0.128	0.097	1.72	62,524
8	1071 sw	1.31	1.93	4.31	0.082	0.060	0.17	62,668
9	1163 nw	2.37	3.36	9.55	0.162	0.105	1.31	62,835
10	1163 se	2.97	4.74	16.52	0.190	0.140	5.41	62,843
11	1165 se	2.07	3.04	8.16	0.130	0.082	1.93	61,874
12	1173 n	1.83	2.63	6.38	0.437	0.091	21.36	58,707
13	1273 n	1.90	2.99	7.41	0.123	0.095	0.83	61,176
14	1276 e	1.56	2.24	6.85	0.154	0.091	1.04	60,057
15	1276 w	3.52	5.50	27.63	0.271	0.197	12.92	59,959
16	1385 n	2.10	3.24	11.30	0.190	0.149	6.18	58,161
17	1404 e	2.69	4.38	11.19	0.201	0.136	6.28	61,257
18	1404 w	2.60	3.93	10.09	0.216	0.120	6.53	60,197
19	1407 n	1.90	3.20	10.03	0.129	0.101	1.15	62,645
20	1407 s	2.14	3.12	9.71	0.171	0.109	3.80	62,573
21	1413 ne	1.58	2.27	6.06	0.102	0.076	0.34	62,277
22	1413 sw	1.64	2.25	5.92	0.100	0.076	0.11	62,174
23	1414 ne	1.54	2.13	6.18	0.124	0.081	0.39	60,300

#	Sensor	MAE (mph)	RMSE (mph)	MAPE (%)	Speed CoV	Within-slot CoV	Congestion (%)	Test samples
24	1414 sw	1.41	1.98	5.66	0.114	0.072	0.29	60,317
25	1417 s	1.47	2.08	7.07	0.111	0.091	0.20	62,237
26	1418 e	1.92	3.28	9.09	0.118	0.096	1.44	55,437
27	1418 w	1.74	3.06	7.90	0.093	0.090	0.57	55,694
28	1420 nw	2.34	3.59	10.31	0.158	0.109	3.37	62,082
29	1420 se	1.91	2.90	8.64	0.128	0.106	0.92	61,899
30	1425 e	1.80	2.48	3.36	0.066	0.048	0.14	62,208
31	1425 w	2.16	4.38	6.25	0.080	0.074	0.51	62,376
32	1426 e	2.26	3.25	4.32	0.075	0.066	0.17	61,908
33	1426 w	3.75	6.26	12.68	0.171	0.121	3.26	62,685
34	1428 e	3.88	7.14	12.68	0.164	0.113	5.04	61,259
35	1428 w	2.17	3.25	4.12	0.066	0.060	0.06	61,158
36	1429 e	2.02	2.81	7.04	0.108	0.088	0.07	61,680
37	1429 w	2.00	2.90	7.04	0.108	0.095	0.25	60,874

Aggregate: mean MAE 2.07 mph (median 1.92); mean MAPE 8.47% (median 7.41%). The highest-error sensor directions on this test period are 1276 west (MAPE 27.6%) and 1163 southeast (16.5%). Per-sensor error correlates strongly with the two series properties in the table: on this test period, Spearman rho = 0.78 between MAPE and Speed CoV, and rho = 0.80 between MAPE and congestion frequency (rho = 0.79 and 0.82 respectively for the 4.2-year cold-start regime evaluated on its own 2024-2025 held-out period, so the relationship is stable across both training regimes and test periods). The relationship is a strong rank trend rather than a per-sensor rule, and the exceptions are informative. Sensor 1173 north has the highest Speed CoV (0.437) yet only mid-range MAPE (6.4%): its whole-series CoV is dominated by a historical level shift (yearly mean speed 8.4, 14.3, 47.4, 46.8 km/h for 2020-2023), and in its stable regime it is highly predictable (a time-of-week average over the last four months of training data already achieves 5.9% MAPE). Conversely, 1276 west has stable yearly levels and a strong weekly rhythm, but large day-to-day spread within the same time-of-week slot (a time-of-week average still leaves 22% MAPE) - variation that no calendar-feature model can capture, which is what its 27.6% error reflects. This motivates a sharper variability measure: the within-time-slot coefficient of variation (standard deviation of speed after removing each sensor's time-of-week mean profile, divided by mean speed), computed on a stable year of data (2023). Against this measure the correlation rises to Spearman rho = 0.94, and the exceptions dissolve: 1276 west ranks first on both within-slot CoV (0.197) and MAPE, while 1173 north falls to rank 21 of 37, matching its error rank. In short, prediction difficulty is driven almost entirely by day-to-day spread at the same time-of-week; whole-series CoV is an imperfect proxy because it also counts historical level shifts, which a year-aware model absorbs. Note that the error ranking is period-dependent: on the 2024-2025 common-test-set windows (next section), platform 1404 becomes the dominant outlier (MAPE up to 86.5%). The sensor data shows why: both directions slowed progressively from ~57 km/h (August 2024) to ~33 km/h (January 2025) with variance roughly tripling - a months-long, bidirectional degradation consistent with long-term roadworks at the site - so models trained before the change are evaluated against a changed road.

Common-test-set scenario comparison

All training scenarios of the paper (3-year base, 4.2-year cold start, and progressive warm-start re-training windows PW3/PW6/PW12) were additionally evaluated on identical held-out periods. Mean MAE (mph) per evaluation window:

Scenario	Dec 2024 - Feb 2025	Oct 2024 - Feb 2025	May 2024 - Feb 2025
base (3y)	3.09	2.83	2.44
cold start (4.2y)	3.02	2.71	2.32
re-train PW3	2.91	2.68	2.35
re-train PW6	2.89	2.64	2.31
re-train PW12	2.90	2.61	excluded*

*PW12 trains up to September 2024, overlapping the nine-month window.

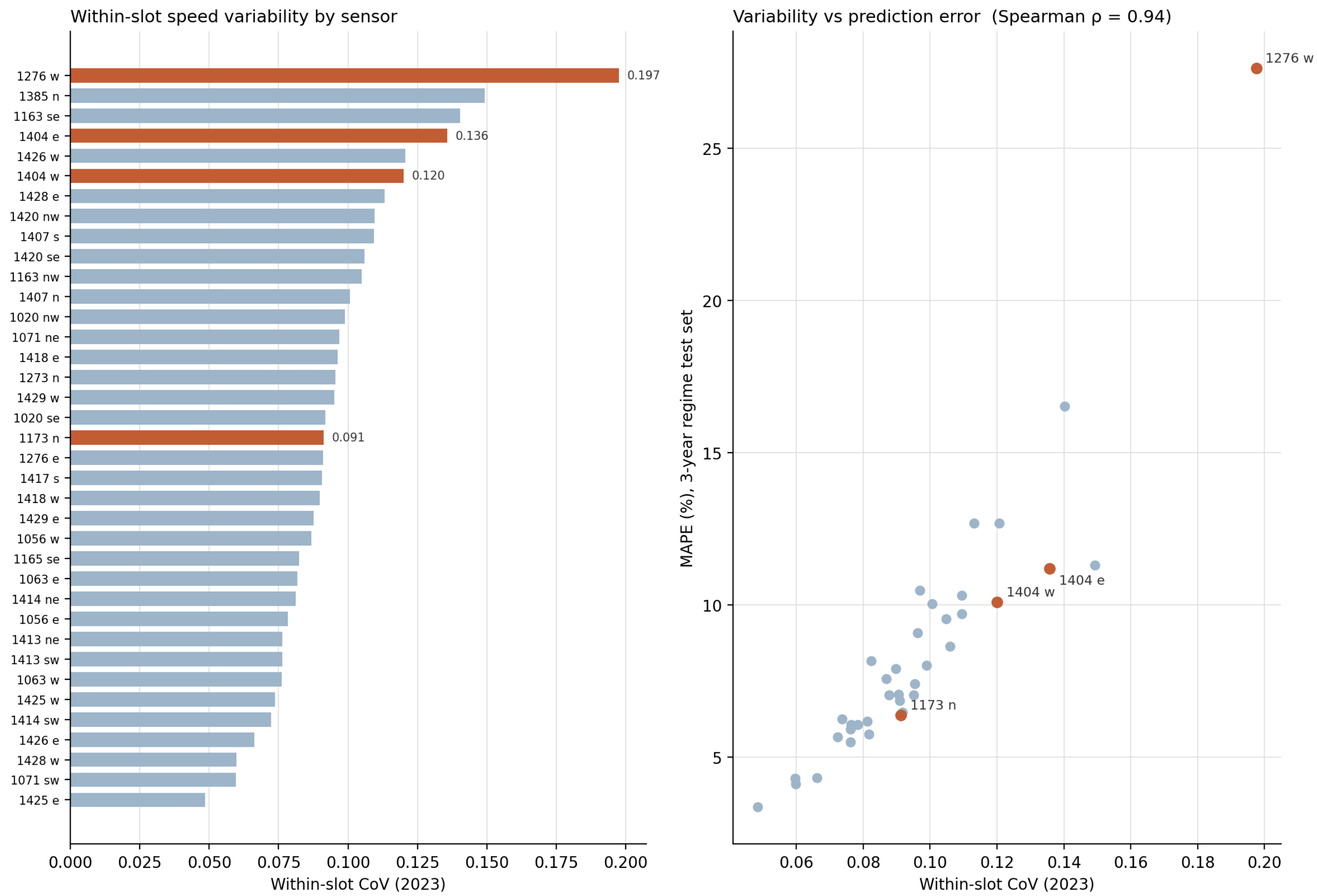

Figure 1: Left: within-slot CoV (2023) per sensor, sorted; highlighted bars are the sensors discussed in the text. Right: within-slot CoV against held-out MAPE under the 3-year regime (Spearman rho = 0.94).

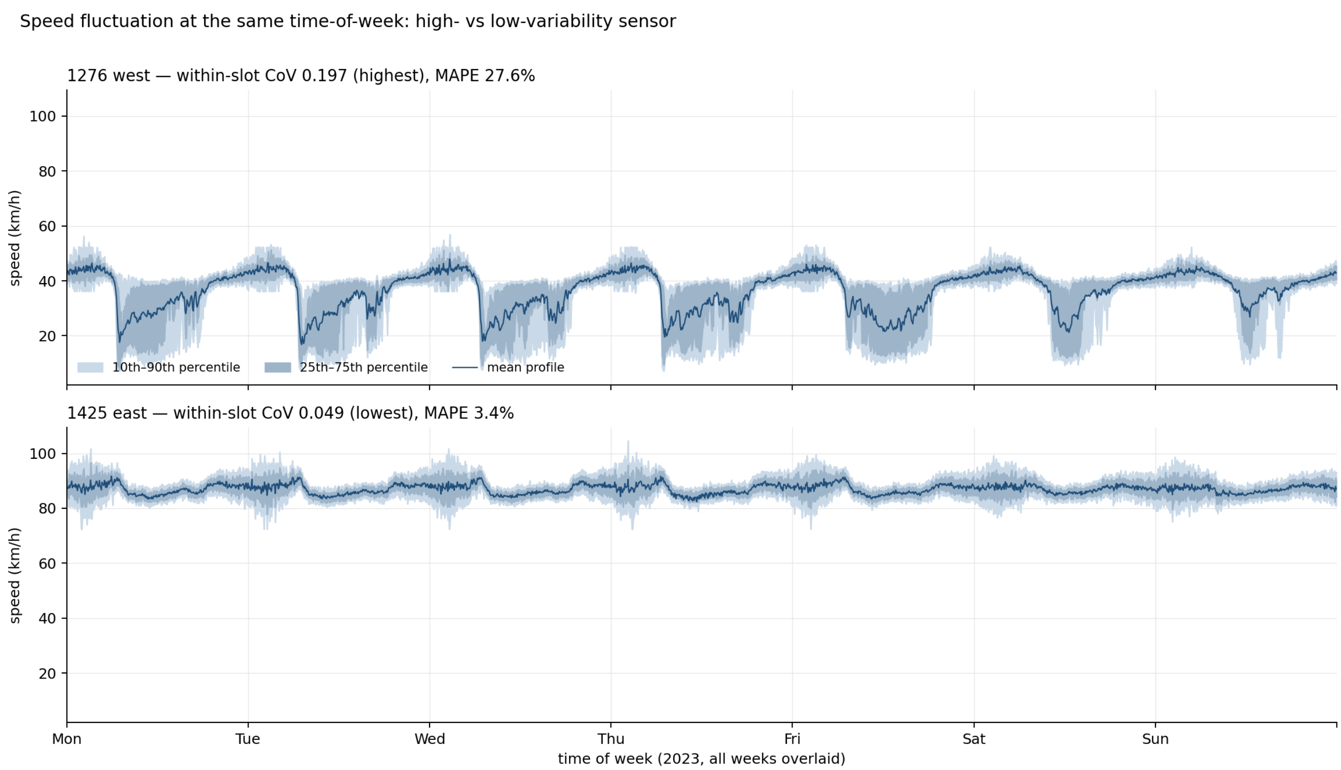

Figure 2: The within-slot spread seen directly: all weeks of 2023 folded onto one Monday-Sunday axis at 5-minute resolution. At each slot, the line is the mean across the ~52 weeks and the bands the 25th-75th and 10th-90th percentiles of those same weeks. On 1276 west (top) the same clock time spans near-standstill to free flow depending on the week; on 1425 east (bottom, lowest within-slot CoV) the band stays a narrow ribbon. A calendar-feature model predicts one value per slot, so the band width bounds its achievable accuracy.

Warm-start re-training improves on both cold-start baselines in every window; prediction accuracy is invariant to the training platform (across repeated runs and platforms, the standard deviation of aggregate MAPE is below 0.3 percentage points in over 90% of measured cells).